ENTERPRISE VIDEO SURVEILLANCE · SYSTEM ARCHITECTURE

HikCentral Professional VMS Network Architecture

500 × 4 MP IP-camera deployment — end-to-end network topology, device model specification, VLAN segmentation and fibre backbone design.

≈500 4 MP Cameras | **6** Access Switches | **3** Distribution | **2** Redundant Core | **25G** Backbone | **10** Workstations

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LAYER 1 → LAYER 6

End-to-End Network Topology

Left → Right signal & data flow



◇ DATA CENTRE / SERVER FARM — 25G ATTACHED

×1
HikCentral Pro
Mgmt Server (SYS)
P-VER / HW1D

×5
Recording Servers
Hybrid SAN

×3
Streaming Servers
HCP Stream role

×5
Storage — Hybrid SAN
DS-A81024S

◇ OPERATOR CLIENTS

VMS Workstations
Intel Core i7 · dual 4K

WS1 WS2 WS3 WS4 WS5 WS6 WS7
WS8 WS9 10

BILL OF MATERIALS

Device Models & Specifications

Recommended enterprise-grade hardware

MANAGEMENT SERVER (SYS / CMS) ×1

HikCentral Pro Server

HIKCENTRAL-P-VER / HW1D

- 1U rack appliance, Xeon, RAID-1, iDRAC9
- Windows Server 2019, HCP pre-licensed
- + RoseReplicator Plus hot-standby (HA)

RECORDING SERVERS ×5

Hybrid SAN (CVR)

DS-A81024S

- Added into HCP as Recording Servers
- N+1 hot-spare & video copy-back
- ANR fill-in for network gaps

STORAGE — HYBRID SAN ×5

24-Bay SAN Array

DS-A81024S

- 24 × 8 TB SATA · dual RAID-5
- ≈160 TB raw / ≈151 TB usable each
- Dual controller · dual PSU · iSCSI

STREAMING SERVERS ×3

HCP Streaming Role

1U Xeon Server

- Offloads live-view & playback fan-out
- Protects recording-path bandwidth
- Serves the 10 operator workstations

CORE SWITCHES ×2

Cisco Catalyst 9500

C9500-24Y4C

- 24 × 25G SFP28 + 4 × 100G QSFP28
- StackWise Virtual redundant pair
- UADP 3.0 ASIC · L3 routed core

SERVER-FARM SWITCH ×1

Cisco Catalyst 9500

C9500-48Y4C

- 48 × 25G SFP28 + 4 × 100G QSFP28
- Low-latency server & SAN fabric
- OM4 multimode to core

DISTRIBUTION SWITCHES ×3

Cisco Catalyst 9300X

C9300X-24Y

- 48 × 1/10/25G · modular uplinks
- StackWise-1T · zone aggregation
- 25G up to core / 10G down to access

ACCESS SWITCHES ×6

Cisco Catalyst 9300

C9300-48P

- 48 × 1G PoE+ (437 W) for cameras
- 10G fibre uplinks (OS2) to distribution
- ≈85 cameras / switch · spare capacity

VMS CLIENT WORKSTATIONS ×10

Control Client PC

Intel Core i7

- Core i7 (13/14th gen) · 32 GB RAM
- Dedicated GPU · dual 4K display
- 1 GbE to nearest access/distribution

ADDRESSING & TRANSPORT

VLAN Segmentation & Fibre Backbone

10.10.0.0 /16 scheme

VLAN & IP PLAN

ZONE	VLAN	SUBNET
Management / Server zone	55	10.10.55.0/24
Distribution 1 — camera zone	51	10.10.51.0/24
Distribution 2 — camera zone	52	10.10.52.0/24
Distribution 3 — camera zone	53	10.10.53.0/24

~167 cameras per /24 zone — well within 254 usable hosts, leaving headroom for growth. Server, recording, streaming & SAN all sit in the isolated management VLAN 55; inter-VLAN routing is handled at the L3 core.

FIBRE & LINK MATRIX

LINK TIER	SPEED	CABLE	OPTIC
Access → Distribution	10G	OS2 · SM	10G-LR
Distribution → Core	25G	OS2 · SM	25G-LR
Core → Server-Farm	25G	OM4 · MM	25G-SR
Core ↔ Core (SVL)	100G	OM4 · MM	100G-SR4

Single-mode (OS2) on the long building runs — access & distribution — for reach and future upgrades. **Multimode (OM4)** for the short core-to-server-farm hops in the equipment room (25G-SR to ~100 m). Match SFP28 optics to each media type.